

Plane-Decomposition Readout in N-Body Fixed-Generator Systems

Each rotation plane reads as kinematically isomorphic to the two-body AB system — plane-wise validity of progression, frequency, amplitude, energy, and the zero-order harmonic structure, with position and observed-acceleration readouts as connection problems

Author: Noriaki Kihara

ORCID: 0009-0004-6753-4020

Date: July 21, 2026

Version DOI: 10.5281/zenodo.21468960

Concept DOI: 10.5281/zenodo.21468959

Position: Dimensional Generation Structure series, Paper 4, v1

Abstract

The preceding Wave Information Readout series established, in the closed two-body AB phase system, position and harmonic readout [2] and the acceleration-like readout with the inverse-square law [3]. Paper 3 [5] proved that the fixed generator of N-body complete pairwise relational waves decomposes into a direct sum of rotation planes and a null space, with the number of rotation modes growing at most linearly. This paper connects the two series and answers the following question.

From which part of an N-body fixed-generator system, and to what extent, do the readouts established for the two-body AB system hold?

The key is that under a fixed generator each rotation plane is an invariant subspace and the motion on it is a single-frequency phase progression. We formalize this as the plane AB kinematic isomorphism and establish five mathematical consequences. First, the phase progression of each plane advances by the constant angle $\theta_j = 2 \arctan(\gamma\sigma_j)$ with constant plane amplitude (Theorem 1). Second, under the in-plane $SO(2)$ gauge (the orientation of each plane is fixed by the generator), the absolute phase is unreadable while the signed phase progression, frequency, and amplitude are readable; if reflections are allowed ($O(2)$), only the absolute values of progression and frequency are invariant. This means that Axiom 5 of the basic axiom system (unreadability of individual phases) is reproduced at the level of derived quantities (Theorem 2). Third, the total energy decomposes exactly into per-plane and kernel conserved quantities: $H = \sum_j H_j + H_{\text{ker}}$ (Theorem 3). Fourth, because the residual subspace (kernel) carries no phase progression, progression-based readouts — position as progressing phase and harmonic acceleration — do not hold from the kernel; the projector, the kernel squared quantity, and the state projection P_0X are covariantly defined, while K alone cannot select a standard internal basis (the dynamical realization of Consequence 16.3 [1], Theorem 4). Whether the static P_0X can be read as a positional direction is not tested in this paper. Fifth, the squared closure itself also decomposes plane-wise and kernel-wise with each term individually conserved: $\sum_j Q_j + Q_{\text{ker}} = R^2$ (Theorem 5). Under a fixed generator, therefore, neither closure nor energy-like quantities move between planes — a no-go condition for spontaneous splitting of the state.

In the numerical experiments ($N = 3-9$, 32 trials $\times$ 720 steps each, fixed generator, no sequential renormalization) we confirmed: plane-amplitude drift $\leq 1.21 \times 10^{-13}$; per-step progression variation $\leq 1.89 \times 10^{-13}$; deviation of the signed mean progression from $+2 \arctan(\gamma\sigma_j) \leq 4.49 \times 10^{-15}$; signed-progression change under the in-plane $SO(2)$ gauge $\leq 1.52 \times 10^{-13}$; deviation of the orientation convention $Kp_j = \sigma_j q_j \leq 7.64 \times 10^{-14}$; deviations of the permuted frequency and energy sets $\leq 3.30 \times 10^{-15}$; frequency nondegeneracy (minimum gap $\geq 2.0 \times 10^{-3}$). The second-difference relation $\Delta^2 c_j = -4 \sin^2(\theta_j/2) c_j$ held with absolute deviation $\leq 1.27 \times 10^{-16}$ (machine precision), and the distance exponent in the continuous-spectrum re-expression, for all configurations with at least two planes ($N = 4-9$), was -1.99999 to -1.99987 . The energy-decomposition identity held to $\leq 2.00 \times 10^{-15}$, the closure-decomposition identity $\sum Q_j + Q_{\text{ker}} - R^2$ to $\leq 1.89 \times 10^{-15}$, the per-plane closure variation to $\leq 1.81 \times 10^{-13}$, the kernel component was static to $\leq 4.57 \times 10^{-14}$, the kernel-projector gauge variation was $\leq 7.77 \times 10^{-16}$, and the angle between independent kernel bases was at least 13.6° .

This is a verification paper: most results were predicted as consequences of the structure of Paper 3. The substantive novelty is twofold. First, the asymmetric structure of readability — on rotation planes, progression, frequency, amplitude, energy, and the zero-order harmonic structure hold through the kinematic isomorphism, while progression-based readouts fail on the residual subspace — is fixed by both theorems and numerics. Second, the fixed generator carries a double conservation register per plane and kernel (H_j and Q_j), and neither quantity moves between planes (the spontaneous-splitting no-go, Theorem 5). The plane counterpart of the AB same-time two-wave positional phase readout (D_{AB}) and the construction of the observed acceleration of Axiom 17 on the planes are not carried out here and are stated explicitly as connection problems.

0. Conclusions

The readability of an N-body fixed-generator system has the following asymmetric structure.

Rotation planes (the 2D blocks of rank K):

progression, frequency, amplitude, energy, and the zero-order harmonic structure all hold (AB kinematic isomorphism)

Residual subspace ($\ker K$):

progression-based readouts fail (static); projector, squared quantities, and state projection are covariantly defined.

Each rotation plane behaves, as a constant-amplitude single-frequency two-dimensional rotation, kinematically isomorphic to the two-body AB system. Phase progression, frequency, amplitude, energy, and the zero-order harmonic second-order structure lift plane-wise to N bodies through this isomorphism. By contrast, the plane counterpart of the AB same-time two-wave positional phase readout (D_{AB}) and the construction on the planes of the observed acceleration of Axiom 17 (curvature radius, future phase position, tangential direction) are not carried out in this paper; they are connection problems.

That under the in-plane $SO(2)$ gauge the absolute phase is unreadable while the signed progression is readable shows that Axiom 5 (unreadability of individual phases) [1] is reproduced not only as an axiom for elementary waves but also at the level of derived quantities obtained by spectral decomposition.

The total energy decomposes as

$$H = \sum_{j=1}^r H_j + H_{\ker}$$

into per-plane and kernel conserved quantities. Moreover, the squared closure itself decomposes as

$$X^T X = \sum_{j=1}^r Q_j + Q_{\ker} = R^2, \quad Q_j = (p_j^T X)^2 + (q_j^T X)^2,$$

with each term individually conserved (Theorem 5). The fixed generator carries a double conservation register per plane and kernel; neither closure nor energy-like quantities move between planes. This is a no-go condition for spontaneous splitting of the state: splitting requires sequentially reconstructed generators, nonlinear couplings, or inter-plane exchange.

The residual subspace is static under the fixed generator and carries no phase progression, so progression-based readouts (position, harmonic acceleration) do not hold from the kernel. The projector, the kernel squared quantity, and the state projection $P_0 X$ are covariantly defined, and the standard internal basis of the kernel cannot be selected from K alone (Consequence 16.3 [1]). Whether the static $P_0 X$ can be read as a positional direction is untested.

Spatial adoption (which planes are read as XYZ) is governed by the adoption conditions of Axiom 16 [1] and is not the subject of verification here. What this paper fixes is the validity range of internal-quantity readout, prior to adoption.

1. Research Question

1.1 Connecting the two series

The Wave Information Readout series established position and harmonic readout [2] and the acceleration-like readout with the inverse-square law [3] in the closed two-body AB phase system. All of these are readouts constructed from the progression of a single phase relation.

The Dimensional Generation Structure series established the structure of N-body complete pairwise relational waves. By Paper 3 [5], the fixed generator satisfies

$$\text{rank } K \leq 2 \min \left(N, \left\lfloor \frac{M}{2} \right\rfloor \right)$$

and decomposes into a direct sum of rotation planes and a null space (Consequences 16.1, 16.2 [1]).

The remaining question was the connection of the two series. The AB readouts read one phase relation; each rotation plane of the N-body system carries one phase progression. Hence the part of the AB readouts that can be constructed from a phase progression should lift plane-wise to N bodies. This paper writes that lift as theorems and verifies it numerically, and at the same time distinguishes the parts whose lift has not yet been derived — the plane counterpart of the same-time two-wave positional phase readout (D_{AB}) and the observed-acceleration projection of Axiom 17 — as connection problems.

1.2 Rule on the direction of causation — plane coordinates are derived quantities

Plane coordinates are not the relational waves themselves; they are derived quantities obtained from the spectral decomposition of the generator. To call a plane readout a readout, one must first show that it can be reconstructed covariantly under name permutations from relational data alone. This is done in Section 3, together with the strict separation of unreadable quantities (absolute in-plane phase, internal kernel directions) from readable ones.

No background space is introduced. No distance matrices, no embeddings, no external observers.

1.3 What this paper discriminates

1. Can each rotation plane be read as a single-frequency rotation kinematically isomorphic to the two-body AB system?
2. Under the in-plane gauge, what is readable and what is not? Is this consistent with Axiom 5?
3. Does the energy-like readout decompose into per-plane and kernel conserved quantities?
4. What can be read from the residual subspace? Do progression-based readouts hold there?
5. Do the zero-order harmonic structure and the continuous-spectrum re-expression exponent hold plane-wise?
6. Is all of the above covariant under name permutations?

2. Classification of Claims

This is a verification paper. All theorems are consequences of the existing structure (the invariant-subspace decomposition of the fixed generator), and the numerical experiments verify their implementation. The axiomatic basis is Basic Axiom System v6 [1].

Item	Classification	Position in this paper
N-body complete pairwise relational-wave model	Model definition	Implementation of Axiom 16 [1]; inherited from [4,5]
Phase-difference sine generator, fixed-generator dynamics	Working hypothesis	Working Axiom I of [1]; identical to [4,5]
Plane coordinates = derived quantities from spectral decomposition	Definition	Section 3; name-permutation covariance is tested

Item	Classification	Position in this paper
Plane AB kinematic isomorphism (constant progression and amplitude)	Mathematical consequence	Theorem 1; does not assert independent plane closure $Q_j = 0$
In-plane $SO(2)$ gauge and readable quantities	Mathematical consequence	Theorem 2; under reflection-including $O(2)$ the signed progression is not invariant
Energy-like decomposition $H = \sum H_j + H_{\text{ker}}$	Mathematical consequence	Theorem 3
Readout restriction on the residual subspace	Mathematical consequence	Theorem 4; dynamical realization of Consequence 16.3
Plane-wise validity of the zero-order harmonic structure	Mathematical consequence (corollary of Theorem 1)	Section 5; distinguished from the observed-acceleration projection of Axiom 17
Plane counterpart of the same-time two-wave positional readout D_{AB}	Connection problem	The corresponding plane quantity is undefined; not constructed here
Plane application of the observed acceleration of Axiom 17	Connection problem	ρ_c , future phase position, and tangential projection are defined after construction on adopted planes
Closure decomposition $\sum Q_j + Q_{\text{ker}} = R^2$ and conservation of each Q_j	Mathematical consequence	Theorem 5
Spontaneous-splitting no-go under fixed generators	Derived consequence	Corollary of Theorem 5; splitting requires dynamics beyond the fixed generator
Continuous-spectrum re-expression exponent -2	Consequence of a definition	Section 5; follows identically from the cell-width definition $\Delta\theta_j = 2\pi\theta_{\text{min}}/\theta_j$
Plane version of integer-harmonic closure (the mechanism of [3])	Not established (unexamined)	Plane frequency ratios are generally non-integer
Frequency nondegeneracy	General-position assumption (independent)	Section 3.3; not derivable from maximal rank; holds numerically in all 224 trials
Results of Experiments F–I	Numerical facts	Section 7
Spatial adoption limited to three planes	Rule of Axiom 16 [1]	Outside the scope of verification here
Behavior of nearly degenerate planes	Out of scope	Treated in the sequel (allocation and observation selection)
First-order readout (velocity feedback) on planes	Out of scope	This paper is restricted to zero order

3. Model and Plane Decomposition

3.1 Inheritance of the model

To the complete pairwise relation set $\mathcal{E}_N$ ($M = \binom{N}{2}$) of N anonymous individuals we assign complex relational waves $X \in \mathbb{C}^M$ with the closure $\sum_e X_e^2 + (iR)^2 = 0$ [1,4]. From the initial phases the phase-difference sine generator $\tilde{K}_{ef} = A_{ef} \sin(\theta_f - \theta_e)$ is constructed once, normalized by its spectral norm, and kept fixed (fixed generator). The update is the real orthogonal Cayley transform

$$U = (I - \gamma K)^{-1}(I + \gamma K), \quad \gamma = \tan(\pi/144),$$

which preserves the squared closure and the absolute-square sum without sequential renormalization [5,8].

3.2 Spectral decomposition and plane coordinates

A real antisymmetric matrix K decomposes under an orthogonal transformation into two-dimensional rotation blocks and a zero block [7]. To each nonzero block j corresponds an orthonormal pair (p_j, q_j) and a frequency $\sigma_j > 0$; to the null space corresponds an orthonormal basis $\{n_a\}$. The plane coordinate is

$$\zeta_j(\tau) = p_j^\top \Re X_\tau + i q_j^\top \Re X_\tau$$

(the same construction runs in parallel for $\Im X_\tau$).

3.3 Well-definedness of the plane readout

The plane readout is grounded as follows. First, (p_j, q_j, σ_j) is determined solely by the spectral decomposition of K constructed from relational data (the initial phases), without external axis names. Second, under a name permutation π , $K' = P_\pi K P_\pi^\top$, so the frequency set $\{\sigma_j\}$ is invariant and the plane bases move covariantly, $p_j \mapsto P_\pi p_j$. The readout quantities — frequencies, amplitudes, energies — are therefore permutation invariant (verified numerically in Experiment F). Third, each plane basis carries a residual gauge. Since K itself fixes the orientation of each plane (the standard form with $\sigma_j > 0$ is fixed by the orientation $K p_j = \sigma_j q_j$), the residual gauge preserving the standard form is $SO(2)$, and by Theorem 2 this freedom does not affect the readout quantities. Allowing reflections ($O(2)$) flips the signed progression.

When frequencies degenerate, additional freedom remains in the very separation into planes. Note that the generic rank equality of Consequence 16.2 [1] guarantees $\sigma_j \neq 0$; $\sigma_j \neq \sigma_k$ ($j \neq k$) is a separate proposition. This paper places frequency nondegeneracy explicitly as an independent general-position assumption. In all 224 trials of the numerical experiments ($N = 3\text{--}9 \times 32$) all frequencies were distinct. The proof that the degeneracy set has measure zero is not given here (Section 8.4). The behavior of near degeneracy is likewise out of scope (Section 8).

4. Plane Readout Theorems

Theorem 1 (plane AB kinematic isomorphism)

Under the fixed-generator dynamics, the coordinate of each rotation plane satisfies

$$\zeta_j(\tau + 1) = e^{+i\theta_j} \zeta_j(\tau), \quad \theta_j = 2 \arctan(\gamma \sigma_j).$$

That is, the phase progression is the constant angle θ_j and the amplitude $|\zeta_j|$ is constant: each plane is, as a constant-amplitude single-frequency two-dimensional rotation, kinematically isomorphic to the rotational kinematics of the two-body AB system [2,3]. The progression readout is defined for planes with nonzero amplitude $|\zeta_j| > 0$ (at $\zeta_j = 0$ the phase is undefined).

Proof. On plane j , K acts as the two-dimensional antisymmetric block

$$\begin{pmatrix} 0 & -\sigma_j \\ \sigma_j & 0 \end{pmatrix},$$

whose Cayley transform is a two-dimensional rotation by $2 \arctan(\gamma \sigma_j)$. The blocks are mutually orthogonal and do not mix, so the plane coordinate rotates at constant angular speed with conserved amplitude. ■

Note that the complex bilinear quadratic quantity $Q_j = (p_j^\top X)^2 + (q_j^\top X)^2$ within the plane is also conserved under the real orthogonal update, but $Q_j = 0$ does not hold. The theorem asserts a kinematic isomorphism, not that each plane is an independently closed subsystem.

Theorem 2 (in-plane $SO(2)$ gauge and readable quantities)

For nonzero amplitude $|\zeta_j| > 0$, under the in-plane basis rotation $(p_j, q_j) \mapsto (p_j \cos \beta + q_j \sin \beta, -p_j \sin \beta + q_j \cos \beta)$ the absolute phase $\arg \zeta_j$ shifts by $-\beta$, while the signed phase progression $\Delta \varphi_j$, the frequency θ_j , and the amplitude $|\zeta_j|$ are invariant. Under the $SO(2)$ gauge, therefore, the uniquely readable quantities are the signed progression, frequency, amplitude, and phase differences; the absolute phase is not readable.

Proof. The basis rotation induces $\zeta_j \mapsto e^{-i\beta} \zeta_j$. The ratio $\zeta_j(\tau+1)/\zeta_j(\tau)$ and the modulus do not depend on $e^{-i\beta}$. ■

The residual gauge is $SO(2)$ because K itself fixes the orientation of the plane (Section 3.3). Under the reflection $(p_j, q_j) \mapsto (p_j, -q_j)$, allowed in $O(2)$, we get $\zeta_j \mapsto \bar{\zeta}_j$ and the signed progression flips, $\Delta\varphi_j \mapsto -\Delta\varphi_j$; the invariants are then $|\Delta\varphi_j|$ and the absolute value of the frequency.

This shows that the unreadability of individual phases, placed as Axiom 5 [1] for elementary waves, is automatically reproduced at the level of the derived plane coordinates. The distinction between readable and unreadable quantities is preserved as one climbs the readout hierarchy.

Theorem 3 (energy-like decomposition)

$$H(X) = \sum_{j=1}^r H_j + H_{\text{ker}}, \quad H_j = |p_j^\dagger X|^2 + |q_j^\dagger X|^2, \quad H_{\text{ker}} = \|P_0 X\|^2$$

holds as an identity, and under the fixed-generator dynamics each H_j and H_{ker} is individually conserved.

Proof. $\{p_j, q_j\} \cup \{n_a\}$ is an orthonormal basis of $\mathbb{R}^M$, so the decomposition is Parseval's identity. U decomposes into rotations within each plane and the identity on the kernel, so each component norm is invariant. ■

Theorem 4 (readout restriction on the residual subspace)

The kernel component $P_0 X_\tau$ does not depend on τ (static). Hence readouts based on nonzero phase progression — position as progressing phase and harmonic acceleration — do not hold from the residual subspace. The projector P_0 , the kernel squared quantity H_{ker} , and the state projection $P_0 X$ are all covariantly defined under name permutations. The standard internal basis of the kernel cannot be selected from K alone.

Proof. If $Kn = 0$ then $Un = n$, so the kernel component is invariant. The progression is zero, so phase increments and second differences vanish identically and progression-based readouts are undefined. Uniqueness of the projector and non-uniqueness of the internal basis follow from Consequence 16.3 [1]. $P_0 X$ is uniquely determined by the state and moves covariantly, $P_0 X \mapsto P_\pi P_0 X$, under permutations. ■

Theorem 4 shows that the uniqueness distinctions stated algebraically by Consequence 16.3 are realized directly as readout availability in the dynamics. Whether the static state projection $P_0 X$ can be read as a “positional direction” is not tested here. The interpretation of Paper 2, which read a one-dimensional kernel as the static Z-normal direction [4], is a precedent located in this untested region and does not contradict the theorem.

Theorem 5 (closure decomposition and splitting no-go)

The squared closure decomposes as

$$X^\top X = \sum_{j=1}^r Q_j + Q_{\text{ker}} = R^2, \quad Q_j = (p_j^\top X)^2 + (q_j^\top X)^2, \quad Q_{\text{ker}} = \sum_a (n_a^\top X)^2,$$

and under the fixed-generator dynamics each Q_j and Q_{ker} is individually conserved.

Proof. Since $\{p_j, q_j\} \cup \{n_a\}$ is an orthonormal basis of $\mathbb{R}^M$, the decomposition is Parseval's identity for the complex bilinear form. For the in-plane rotation $c' = c \cos \theta - d \sin \theta$, $d' = c \sin \theta + d \cos \theta$, the identity $(c')^2 + (d')^2 = c^2 + d^2$ holds algebraically also for complex values c, d ; on the kernel the update is the identity. ■

Corollary (no-go for spontaneous splitting)

Under a fixed generator, neither the closure quantities Q_j nor the energy-like quantities H_j move between planes or between planes and the kernel. Spontaneous splitting of the state accompanied by redistribution among registers therefore cannot occur under fixed-generator dynamics. Splitting requires sequentially reconstructed generators, nonlinear couplings, or inter-plane exchange — dynamics beyond the fixed generator.

Theorem 5 shows that the fixed-generator system carries a double conservation register (H_j and Q_j) per plane and kernel, and serves as the starting point for the sequels on allocation, observation selection, and state splitting.

5. The Zero-Order Harmonic Structure on Planes

By Theorem 1 the plane coordinate rotates by the constant angle θ_j , so the real coordinate $c_j(\tau) = p_j^\top \Re X_\tau$ satisfies the discrete second-difference relation

$$\Delta^2 c_j = -4 \sin^2\left(\frac{\theta_j}{2}\right) c_j$$

(the zero-order harmonic second-order structure). This is a direct corollary of Theorem 1, of the same form as the second-order structure confirmed for the AB system in the inverse-square paper [3].

It must be noted that this is not the observed acceleration of Axiom 17 [1] itself. The readout $\alpha_{\text{read}} = \rho_c |\omega|^2 \hat{\mathbf{t}}_\tau$ of Axiom 17 requires the curvature radius ρ_c , the future phase position, the circumferential direction $\hat{\mathbf{t}}_\tau$, and the projection of the center-directed compensation onto the tangential direction; none of these are constructed on the planes in this paper. Only after Axiom 17 is applied on planes adopted as spatial directions by Axiom 16 can one speak of an observed acceleration. That construction is a connection problem. What this section and Experiment G confirm on all planes is the center-directed zero-order harmonic second-order structure.

To organize the coefficients across planes we define, relative to the slowest plane,

$$n_j = \frac{\theta_j}{\theta_{\min}}, \quad \Delta\theta_j = \frac{2\pi}{n_j} = 2\pi \frac{\theta_{\min}}{\theta_j}.$$

An important caution: n_j is **generally not an integer**. This is therefore a different structure from the integer-harmonic closure $\omega_n = n\omega_1$ ($n \in \mathbb{Z} \setminus \{0\}$) used in the inverse-square paper [3]; it is a re-definition of the cell width on a continuous spectrum (continuous-spectrum re-expression). Under this definition, in the small-angle regime,

$$4 \sin^2\left(\frac{\theta_j}{2}\right) \simeq \theta_j^2 \propto \frac{1}{\Delta\theta_j^2}$$

follows identically. The exponent -2 established here is thus a consequence of the definition; the plane version of the integer-harmonic inverse-square law of [3] is not claimed (unexamined). Moreover, this quantity is the normalized harmonic coefficient $g_j = 4 \sin^2(\theta_j/2)$; the plane acceleration amplitude is $a_j = \rho_j g_j$, depending on the plane amplitude ρ_j . The exponent -2 is the exponent of the coefficient, not of the cross-plane acceleration amplitude.

The second-order structure of this paper is restricted to zero order (fixed generator, frozen ω). The plane application of the first-order readout (velocity feedback, [3] v2) is out of scope and treated in a sequel.

6. The Boundary of Spatial Adoption

What this paper confirms is the readout validity of the **internal quantities** of all r planes — progression, frequency, amplitude, energy, and the zero-order harmonic structure. Which planes are adopted as spatial directions (XYZ) is governed by the adoption conditions of Axiom 16 (distinguishability by rotation frequencies plus normal uniqueness, restricted to $d = 3$) [1], and as Paper 3 showed, adoption saturates at three directions. The experiments here do not verify the adoption limit. That internal quantities are readable on all planes and that spatial adoption is limited to three planes coexist without contradiction — this is the two-layer structure of readout.

7. Numerical Experiments

7.1 Common conditions

Item	Value
Configurations	$N = 3-9$
Trials	32 each
Steps	720 each
Dynamics	Fixed generator, Cayley orthogonal update ($\gamma = \tan(\pi/144)$)
R^2 / imaginary seed amplitude	1.0 / 0.35
Random seed	20260731
Tolerance	10^{-10}
Sequential renormalization / observation damping / absolute background axes	None

Plane counts and kernel dimensions agreed across all 32 trials per configuration, reproducing the rank law of Paper 3, $r = \min(N, \lfloor M/2 \rfloor)$, $\dim \ker K = M - 2r$. No planes were excluded for low amplitude in any trial.

N	M	Rotation planes r	$\dim \ker K$
3	3	1	1
4	6	3	0
5	10	5	0
6	15	6	3
7	21	7	7
8	28	8	12
9	36	9	18

7.2 Experiment F: plane AB kinematic isomorphism

Quantity	Maximum over all configurations
Plane-amplitude drift	1.21×10^{-13}
Per-step progression variation (single-frequency property)	1.89×10^{-13}
Deviation of signed mean progression from $+2 \arctan(\gamma \sigma_j)$	4.49×10^{-15}
Signed-progression change under the in-plane $SO(2)$ gauge	1.52×10^{-13}
Deviation of the orientation convention $(Kp_j) \cdot q_j / \sigma_j - 1$	7.64×10^{-14}
Deviation of the permuted frequency set	2.00×10^{-15}
Deviation of the permuted plane-energy set	3.30×10^{-15}

All claims of Theorems 1 and 2 held three orders below tolerance. Under the in-plane $SO(2)$ gauge the absolute phase moved while the signed progression was invariant. In all 224 trials the plane frequencies were all distinct (minimum frequency gap 2.0×10^{-3} to 5.4×10^{-2}), so the general-position assumption of frequency nondegeneracy held numerically. The rank law (plane count = $\min(N, \lfloor M/2 \rfloor)$, kernel dimension = $M - 2r$) also held in every trial.

Figure 1 shows a representative $N = 6$ trial. The phase progressions of the six planes stay at constant values (order 10^{-2}) while the change of the kernel component stays at machine precision (order 10^{-16}): the asymmetry of readability appears as a fourteen-order separation.

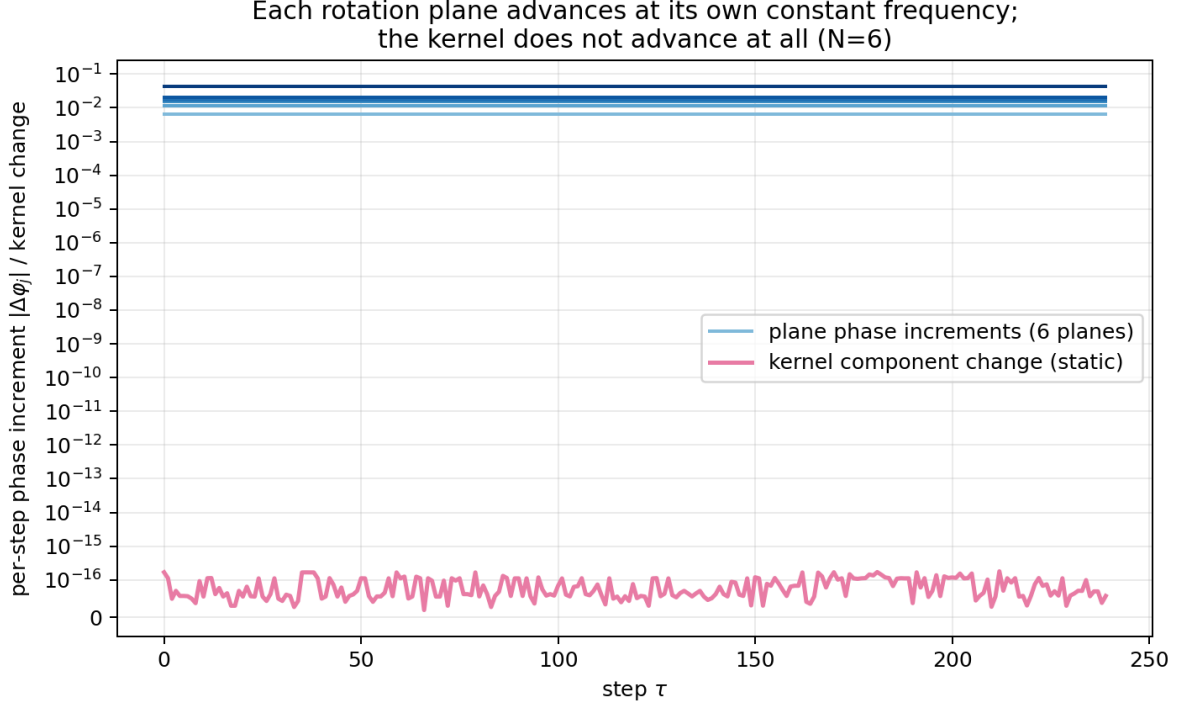


Figure 1: Plane AB kinematic isomorphism and kernel staticity

7.3 Experiment G: plane-wise confirmation of the zero-order harmonic structure

The absolute deviation of the regression slope of $\Delta^2 c_j = -4 \sin^2(\theta_j/2) c_j$ from the theoretical coefficient was at most 1.27×10^{-16} (machine precision) over all configurations. The relative deviation is amplified up to 2.5×10^{-9} on the slowest planes, a conditioning effect of the measure caused by the smallness of the theoretical coefficient itself; the acceptance criterion is the absolute deviation.

The exponent in the continuous-spectrum re-expression, for all configurations with at least two planes ($N = 4-9$) and all trials, was

$$-1.99999 \leq \text{exponent} \leq -1.99987,$$

matching -2 (deviation $\leq 1.2 \times 10^{-4}$, a known discrete effect of $4 \sin^2(\theta/2)$ versus θ^2).

Figure 2 shows the nine planes of a representative $N = 9$ trial lying on the slope -2 line in the continuous-spectrum cell-width display. This visualizes the definitional consequence of Section 5, not a verification of integer-harmonic closure.

7.4 Experiment H: energy-like decomposition and closure decomposition

Quantity	Maximum over all configurations
Per-plane energy H_j drift	2.41×10^{-13}
Kernel energy H_{ker} drift	4.55×10^{-14}
Identity $\sum_j H_j + H_{\text{ker}} - H$	2.00×10^{-15}
Per-plane closure Q_j drift	1.81×10^{-13}
Kernel closure Q_{ker} drift	4.03×10^{-14}
Closure-decomposition identity $\sum_j Q_j + Q_{\text{ker}} - R^2$	1.89×10^{-15}

Figure 3 summarizes the drifts over all configurations. The total energy decomposes exactly into per-plane plus kernel individually conserved quantities. The squared closure R^2 likewise decomposes in the same basis, with each Q_j individually conserved (numerical confirmation of Theorem 5). Under the fixed generator, neither conservation register moves between planes.

7.5 Experiment I: readout restriction on the residual subspace

Quantity	Value
----------	-------

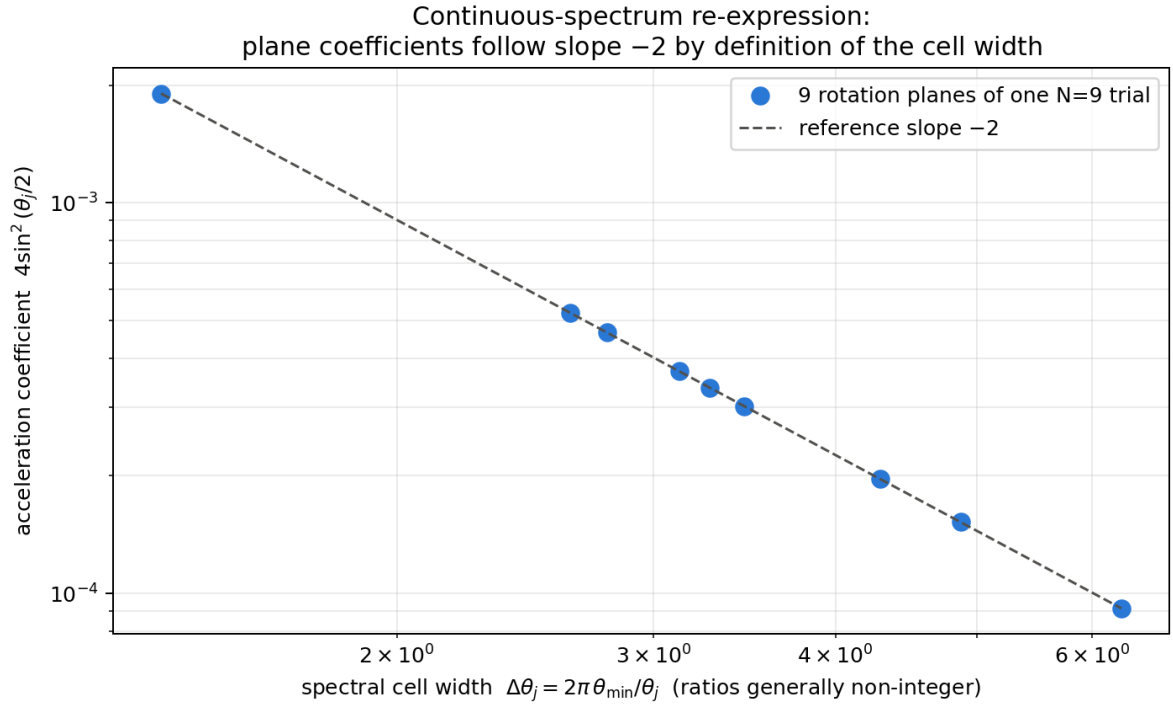


Figure 2: Inverse-square bookkeeping across planes

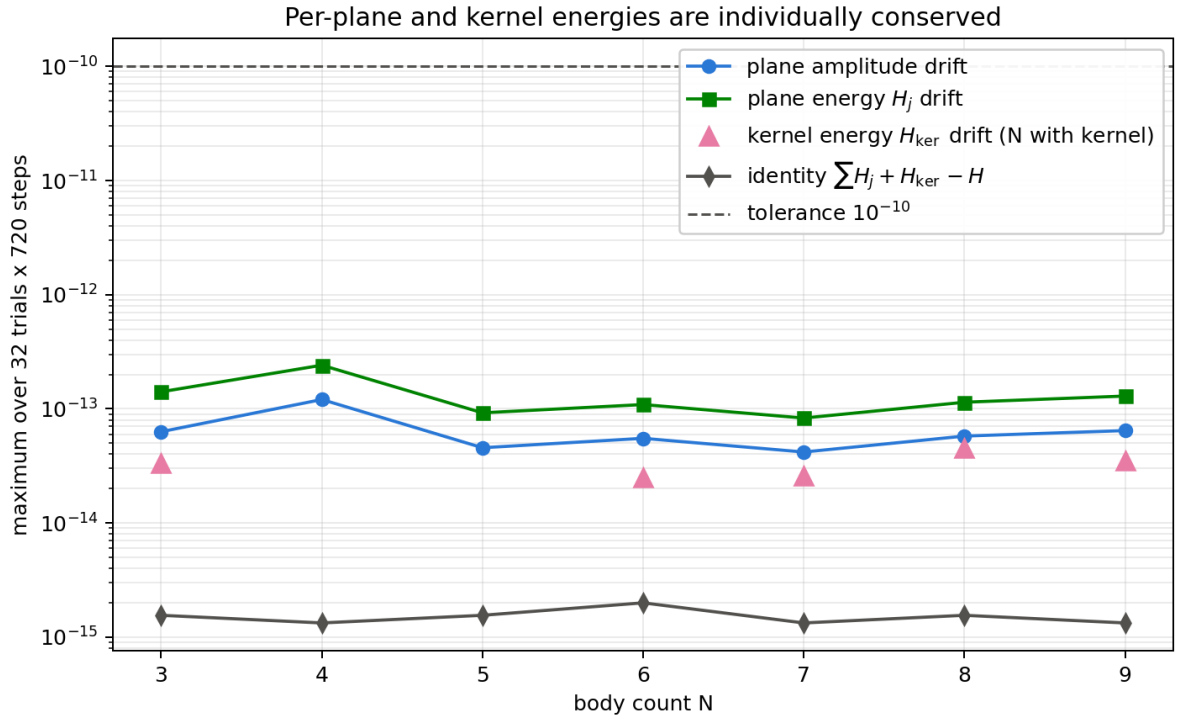


Figure 3: Conservation of the energy-like decomposition

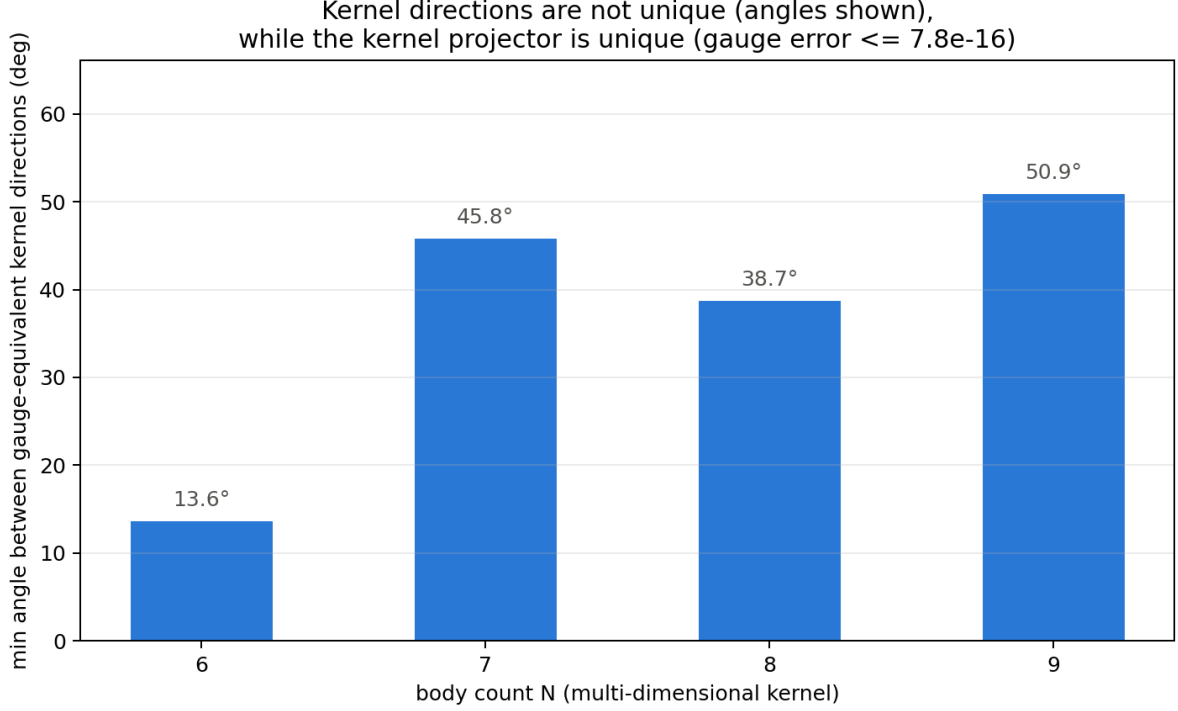


Figure 4: Readout asymmetry of the kernel

2. Under the in-plane $SO(2)$ gauge the absolute phase is unreadable while signed progression, frequency, and amplitude are readable; under $O(2)$ only absolute values (Theorem 2).
3. Energy-like decomposition $H = \sum H_j + H_{\text{ker}}$ with individual conservation (Theorem 3).
4. Readout restriction on the residual subspace: progression-based readouts fail; projector, squared quantities, and state projection are covariant (Theorem 4).
5. Closure decomposition $\sum Q_j + Q_{\text{ker}} = R^2$ with individual conservation, and the spontaneous-splitting no-go (Theorem 5).

8.2 Numerical facts confirmed

1. All claims of Theorems 1–5 held for $N = 3-9$, 32 trials each, within tolerance 10^{-10} (measured values mostly below 10^{-13}).
2. The readout quantities (frequency and energy sets) were permutation invariant. The orientation convention $Kp_j = \sigma_j q_j$ and frequency nondegeneracy (an independent general-position assumption) held in all trials.
3. The zero-order harmonic structure $\Delta^2 c_j = -4 \sin^2(\theta_j/2) c_j$ held at machine precision, and the continuous-spectrum re-expression exponent matched -2 ($N = 4-9$).
4. The rank law and kernel dimensions reproduced the results of Paper 3.
5. The individual conservation of Q_j , Q_{ker} and the closure-decomposition identity held at machine precision (numerical confirmation of Theorem 5).

8.3 Interpretations adopted

1. Axiom 5 (unreadability of individual phases) is reproduced not only for elementary waves but also at the level of derived plane coordinates.
2. The readout asymmetry of Consequence 16.3 (projector and squared quantities unique; internal directions not) is realized directly as readout availability in the dynamics.
3. Readout has a two-layer structure: internal-quantity readout holds on all planes, while spatial adoption is limited to three planes by Axiom 16.

8.4 What this paper does not fix

1. Behavior of nearly degenerate planes; treated in the sequel (allocation and observation selection).
 2. Proof that the frequency-degeneracy set has measure zero; nondegeneracy is placed as an independent general-position assumption.
 3. Whether the plane version of integer-harmonic closure (the inverse-square mechanism of [3]) holds; the exponent -2 here is a definitional consequence of the continuous-spectrum re-expression.
 4. Whether the static state projection P_0X can be read as a positional direction (extension of the Z-normal interpretation of the one-dimensional kernel [4] to multi-dimensional kernels).
 5. Definition and verification of a plane quantity corresponding to the same-time two-wave positional phase readout of the AB system ($D_{AB} = \{\delta, 2\pi - \delta\}$).
 6. Construction and verification of the observed acceleration of Axiom 17 (ρ_c , future phase position, tangential projection) on adopted planes.
 7. Plane application of the first-order readout (velocity feedback); this paper is restricted to zero order.
 8. Dynamical verification of the spatial adoption limit (three planes); the experiments concern internal-quantity readout only.
 9. Stability of the plane structure under sequentially reconstructed generators.
 10. Inter-plane interactions (Group B: exchange scattering).
-

9. Reproducibility

The numerical experiments are based on the following programs and outputs.

Programs

- `dimension_generation_structure/contrast_experiments_v1/run_nbody_plane_decomposition_readout_pre` (Experiments F–I)
- `dimension_generation_structure/contrast_experiments_v1/make_nbody_plane_decomposition_figures_v1` (figures)

Main outputs

- `plane_decomposition_readout_result_v1.json`
- `plane_decomposition_trials_v1.csv`
- `plane_decomposition_readout_report_v1.md`
- Four figures (`plane_ab_isomorphism_v1.png` etc.)

Stored under

`dimension_generation_structure/figs/`.

10. Final Conclusions

The N-body fixed generator decomposes into independent single-frequency rotation planes and carries per-plane and kernel conservation registers. Each rotation plane is, as a constant-amplitude single-frequency two-dimensional rotation, kinematically isomorphic to the rotational kinematics of AB, lifting phase progression, frequency, amplitude, energy, and the zero-order harmonic second-order structure. The plane counterpart of the same-time two-wave positional readout (D_{AB}) and the construction on planes of the observed-acceleration projection of Axiom 17 remain as connection problems.

The total energy decomposes exactly into per-plane and kernel conserved quantities. The residual subspace carries no phase progression, so progression-based position and harmonic-acceleration readouts do not hold there; the projector, kernel squared quantity, and state projection P_0X are covariantly defined, and whether the static P_0X reads as a positional direction is untested.

The conclusions of this paper are therefore:

the N-body fixed generator decomposes into independent single-frequency rotation planes, carries a double conservation register (H_j and Q_j) per plane and kernel, and progression-based readouts fail on the residual subspace.

This asymmetric readability shows that the uniqueness distinctions given algebraically by Consequence 16.3 become readout availability in the dynamics. The no-go of Theorem 5 — under a fixed generator neither closure nor energy-like quantities move between planes — means that spontaneous splitting of the state requires dynamics beyond the fixed generator (sequential reconstruction, nonlinear coupling, or inter-plane exchange). The double registers $\{H_j, Q_j\}$ and $\{H_{\text{ker}}, Q_{\text{ker}}\}$ are the foundation for the sequels on N-body observation selection (weak C readout, strong D observation) and state splitting.

References

Self-citations

1. Noriaki Kihara, “Anonymous Equal-Amplitude Composite-Wave Model: Basic Axiom System v6,” Zenodo, 2026. Version DOI: 10.5281/zenodo.21465984, Concept DOI: 10.5281/zenodo.21315735.
2. Noriaki Kihara, “Harmonic Readout and $c=1$ Area-Sweep Preliminary Experiments in the Closed AB Two-Body Phase System, v4,” Zenodo, 2026. Version DOI: 10.5281/zenodo.21374317, Concept DOI: 10.5281/zenodo.21318696.
3. Noriaki Kihara, “An Inverse-Square Law from a Future-Phase-Position Acceleration Map and Harmonic Closure in a Closed Two-Body AB Phase System, v4,” Zenodo, 2026. Version DOI: 10.5281/zenodo.21468270, Concept DOI: 10.5281/zenodo.21441081.
4. Noriaki Kihara, “Three XYZ Directions Read from Complete Pairwise Relational Waves: Growth of Internal Relational Directions and Saturation of Spatial-Direction Readout in Closed AB, ABC, and ABCD Systems,” Zenodo, 2026. Version DOI: 10.5281/zenodo.21454790, Concept DOI: 10.5281/zenodo.21454789.
5. Noriaki Kihara, “Linear Upper Bound on Generator Rank and Three-Direction Saturation of Spatial-Direction Readout in N-Body Complete Pairwise Relational Waves,” Zenodo, 2026. Version DOI: 10.5281/zenodo.21465899, Concept DOI: 10.5281/zenodo.21465898.

External references

6. Hassler Whitney, “Congruent Graphs and the Connectivity of Graphs,” *American Journal of Mathematics*, 54(1), 150–168, 1932. DOI: 10.2307/2371086.
7. Milan Vujivčić, Fedor Herbut, and Gradimir Vujivčić, “Canonical Form for Matrices Under Unitary Congruence Transformations. I: Conjugate-Normal Matrices,” *SIAM Journal on Applied Mathematics*, 23(2), 225–238, 1972. DOI: 10.1137/0123025.
8. Fasma Diele, Luciano Lopez, and R. Peluso, “The Cayley Transform in the Numerical Solution of Unitary Differential Systems,” *Advances in Computational Mathematics*, 8(4), 317–334, 1998. DOI: 10.1023/A:1018908700358.